MS® ISO 21001: 2025 Certified LATEEFI 40 A CSR Initiative

MS LATEEFI 40 ENTRANCE TEST – 2027

Where Potential Meets Opportunity

For class 10 students

Selection+ Scholarship+ a two year Academic Journey

SYLLABUS FOR EXAMINATION

MATHEMATICS $(20 \times 4 = 80M)$

Chapter / Topic	Syllabus / Topics Covered
Chapter 1: Real Numbers	Introduction, Fundamental Theorem of Arithmetic, Revisiting Irrational numbers, Revisiting Rational Numbers and Their Decimal Expansion.
Chapter 2: Polynomials	Introduction, Geometrical Meaning of the Zeroes of a Polynomial, Relationship between Zeroes and Coefficients of a Polynomial.
Chapter 3: Pair of Linear Equations in Two Variables	Introduction, Pair of Linear Equation in Two variables, Graphical Method of solution of a Pair of Linear Equation, Algebraic Methods of solving a Pair of Linear Equation, Substitution Method, elimination Method.
Chapter 4: Quadratic Equations	Introduction, Quadratic Equations, Solutions of a Quadratic Equation by Factorisation and by using quadratic formula, Nature of Roots.
Chapter 5: Arithmetic Progression	Introduction, Arithmetic Progression, nth Term of an AP, Sum of First n Terms of an AP.
Chapter 6: Triangles	Introduction, Similar Figures, Similarity of Triangles, Criteria for Similarity of Triangles.
Chapter 7: Coordinate Geometry	Introduction, Distance Formula, Section Formula.
Chapter 8: Introduction to Trigonometry	Introduction, Trigonometric Ratios, Trigonometric Ratios of some Specific angles $(0°, 30°, 45°, 60°, 90°)$. Trigonometric Identities.
Chapter 9: Some Applications of Trigonometry	Introduction, problems on Heights and Distances.
Chapter 10: Circles	Introduction, Tangent and Secant of a circle, Number of Tangents from a Point on a Circle.
Chapter 11: Areas Related to Circles	Introduction, Perimeter and Area of a Circle – A Review, Areas of Sector and Segment of a Circle, Areas of combination of Plane Figures.
Chapter 12: Surface Areas and Volumes	Introduction, surface Area of a Combination of Solids, Volume of a Combination of solids.
Chapter 13: Statistics	Introduction, Mean of Grouped Data, Mode of Grouped Data, Median of Grouped data.
Chapter 14: Probability	Introduction, classical definition of probability, problems on finding the probability of an event.

PHYSICS $(15 \times 4 = 60M)$

Chapter / Topic	Syllabus / Topics Covered
Chapter 1: Motion	Distance and displacement, velocity, uniform and non-uniform motion along a straight line; acceleration, distance-time and velocity-time graphs for uniform motion and uniformly accelerated motion, elementary idea of uniform circular motion.
Chapter 2: Force and Laws of Motion	Force and Motion, Newton's Laws of Motion, Action and Reaction forces, Inertia of a body, Inertia and mass, Momentum, Force and Acceleration.
Chapter 3: Gravitation	Gravitation; Universal Law of Gravitation, Force of Gravitation of the Earth (gravity), Acceleration due to Gravity; Mass and Weight; Free fall.
Chapter 4: Work and Energy	Work done by a Force, Energy, Power; Kinetic and Potential energy; Law of conservation of energy (excluding commercial unit of Energy), Transformation of Energy.
Chapter 5: Sound	Nature of sound and its propagation in various media, speed of sound, range of hearing in humans, ultrasound, reflection of sound, echo.

Chapter / Topic	Syllabus / Topics Covered
Chapter 6: Light (Reflection and Refraction)	Reflection of light by curved surfaces; Images formed by spherical mirrors, centre of curvature, principal axis, principal focus, focal length, mirror formula (Derivation not required), magnification. Refraction; Laws of refraction, refractive index. Refraction of light by spherical lens; Image formed by spherical lenses; Lens formula (Derivation not required); Magnification. Power of a lens. Applications of spherical mirrors and lenses.
Chapter 7: Human Eye and the Colourful World	Structure & Functioning of a lens in human eye, defects of vision and their corrections, Refraction of light through a prism, dispersion of light, scattering of light, applications in daily life (excluding colour of the sun at sunrise and sunset).
Chapter 8: Electric Current	Electric current, potential difference and electric current, Ohm's law; Resistance, Resistivity, Factors on which the resistance of a conductor depends. Series combination of resistors, parallel combination of resistors and its applications in daily life. Heating effect of electric current and its applications in daily life.
Chapter 9: Magnetic Effects of Current	Magnetic field, field lines, field due to a current carrying conductor, field due to current carrying coil or solenoid; Force on current carrying conductor, Fleming's Left Hand Rule, Direct current. Alternating current: frequency of AC, Advantage of AC over DC, Domestic electric circuits, Applications of Magnetic Effects of Current.

CHEMISTRY $(15 \times 4 = 60M)$

Chapter / Topic	Syllabus / Topics Covered
Chapter 1: Chemical Equations & Reactions	Modern Symbols of Elements, Chemical Formula, Chemical Equation, Molecular Mass and Mole Concept, Chemical Reactions- Chemical Combination, Decomposition, Displacement, Oxidation & Reduction Reactions.
Chapter 2: Acids, Bases and Salts	Chemical Properties of Acids and Bases, What is Common in All Acids?, Do All Acids Have the Same Strength? Do all bases have the same strength? Acidic & Basic Salts, pH scale.
Chapter 3: Structure of Atoms	Charged Particles in Matter, Atomic Theories (Daltons, Rutherford, Bohr). Distribution of Electrons in Shells, Electronic Configuration.
Chapter 4: Classification of elements	Early Attempts at the Classification of Elements, Modern Periodic Table, Trends on the Basis of Modern Periodic Table
Chapter 5: Metallurgy	Physical Properties of Metals and Non-Metals, Chemical Properties of Metals & Non-Metals, Occurrence of Metals, Extraction of Metals (Metallurgy)
Chapter 6: Carbon and its compounds	Lewis Dot structure of elements (electron dot), Formation of Ionic & Covalent bond. Allotropes of Carbon, Catenation - Unique Property of Carbon, Saturated & Unsaturated carbon compounds. Isomerism-Chain, Branched & Ring. Petroleum (Mineral Oil), Some Compounds of Carbon Other Than Hydrocarbons, Nomenclature of Carbon Compounds, Chemical Properties of Carbon Compounds, Properties of Ethanoic Acid and Ethanol, Saponification (Basic Concepts).
Chapter 7: Atoms and molecules	Symbols of Elements, Valencies, Atomicity, Molecular Mass, Molecular formulae (Criss-cross method), How big are the atoms and molecules?, Chemical Formulae.

BIOLOGY $(10 \times 4 = 40M)$

Chapter / Topic	Syllabus / Topics Covered
Chapter 1: How do Organisms Reproduce	Types of reproduction, Modes of asexual reproduction, Sexual reproduction in plants, Sexual reproduction in animals, Reproduction in humans, Male and female reproductive system, Gestation period, Reproductive health.
Chapter 2: Control and coordination	Control and coordination, Plant hormones, Response to stimulus in plants, Nervous system: Human nervous system, Hormones in animals, The endocrine system, Feedback mechanism, Different control centres in the nervous system, Mechanism of reflex action.
Chapter 3: Heredity and evolution	Heredity, Mendel's Experiment, Sex determination, Evolution, Evidences for evolution, Lamarckism, Darwin's theory, Speciation, Human evolution, Chromosomal disorders
Chapter 4: Respiration – a life process	Discovery of gases involved in respiration, Respiration in Plants, Respiration through roots, Fermentation, Transportation of gases in plants, Stages of respiration in humans, Exchange of gases, Cellular respiration.
Chapter 5: Photosynthesis & Transportation in Plants	Photosynthesis in Plants, Raw material required for Photosynthesis, Mechanism of Photosynthesis. Transportation in Plants – mechanism of water transportation, Transportation of minerals and food materials in plants.

Chapter / Topic	Syllabus / Topics Covered
Chapter 6: Circulation in animals	Human circulatory system, Internal structure of heart, Blood vessels and blood transport, Cardiac cycle, Types of circulation, Lymphatic system, Evolution of transport system, Blood pressure, Blood clotting.
Chapter 7: Excretion	Excretion in human beings, Excretory system, Formation of urine, Urine composition, Dialysis, Kidney transplantation, Accessory excretory organs in humans, Excretion in plants (Alkaloids, tannins, resins, gums, latex)

ENGLISH $(5 \times 4 = 20M)$

Chapter / Topic	Syllabus / Topics Covered
Unseen Comprehension Passage	

MENTAL ABILITY $(15 \times 4 = 60M)$

Chapter / Topic	Chapter / Topic
Chapter 1: Mathematical Operations	**Chapter 6: Puzzles**
Chapter 2: Blood relation	**Chapter 7: Mirror and water images**
Chapter 3: Number series	**Chapter 8: Cubes and Dice**
Chapter 4: Direction sense	**Chapter 9: Ranking order**
Chapter 5: Seating arrangement	**Chapter 10: Coding and Decoding**

SOCIAL SCIENCE $(10 \times 4 = 40M)$

Chapter / Topic	Syllabus / Topics Covered
CIVICS – Chapter 1: Indian constitution	Making of The Indian Constitution, Fundamental Rights
CIVICS – Chapter 2: Democracy	What is Democracy? Features of Democracy, Why Democracy?
CIVICS – Chapter 3: Election Process in India	Election Commission of India, Chief Election Commissioner; The need for Electoral reforms, Political parties -National & State parties.
ECONOMICS – Chapter 4: Sectors of Indian Economy	Sectors of Economic activities, Comparing the three sectors, Primary, Secondary and Tertiary sectors in India. Division of sectors: organized and unorganized, Sectors in terms of ownership: Public and Private sectors.
ECONOMICS – Chapter 5: Money and Credit	Barter System, Reserve Bank of India & Self Help Groups, Modern forms of money, Loan activities of Banks.
ECONOMICS – Chapter 6: Globalization and the Indian Economy	What is Globalization? Factors that have enabled Globalization, World Trade Organization, and Impact of Globalization in India, International Monetary Fund & MNCs.

DEENIYAT $(10 \times 4 = 40M)$

Chapter / Topic	Syllabus / Topics Covered
Chapter 1: Five Pillars of Islam & Months of Islam	**Chapter 4: Family & Important Holy Companions (Names) of Prophet Muhammad PBUH**
Chapter 2: Books of Allah SWT (Prophets & their book names) The Quran (Parahs & Surahs)	**Chapter 5: Angels of Allah SWT (Names & their duties)**
Chapter 3: Prophets of Allah SWT (Names & their miracles)	**Chapter 6: Important milestones in Islamic History**